

STUDENT CONTEST IC-MNSS 2024

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Team: SightSync

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Abstract

Introduction

Visual impairment poses significant challenges to individuals in navigating their surroundings independently. According to a study, currently, there are an estimated 4.95 million blind persons and 70 million vision impaired persons in India. In response to this, **our project aims to develop "SightSync: Affordable Depth Perception and Direction Guiding Glasses for the Visually Impaired with Object Recognition using Computer Vision." These glasses leverage innovative technologies, including depth sensors or computer vision (CV), Arduino microcontrollers, and voice signaling which can be further integrated with Google Maps, to enhance mobility and independence for those with visual impairments .**



Objectives

1. Our primary objectives include designing and implementing depth perception glasses equipped with either LiDAR sensors or a camera for depth perception.
2. We integrate Arduino microcontrollers for real-time data processing and control, employing computer vision-based object detection to identify nearby objects.
3. Voice signaling is implemented to provide users with intuitive feedback and object descriptions.
4. These glasses can be integrated with Google maps so that the user can know the directions for their destination using a virtual assistant. This will help them to make more informed decisions about the path they are traversing.

TIMELINE PLANNED

Week 1 :

- Research and analyze LiDAR sensors and camera modules for depth perception.
- Begin initial designs for depth perception glasses.
- Research computer vision algorithms suitable for object detection.

Week 2 :

- Finalize selection between LiDAR sensors and camera modules for depth perception.
- Begin procurement of necessary components (sensors, Arduino microcontrollers, etc.).
- Continue research on computer vision algorithms and libraries.

Week 3 :

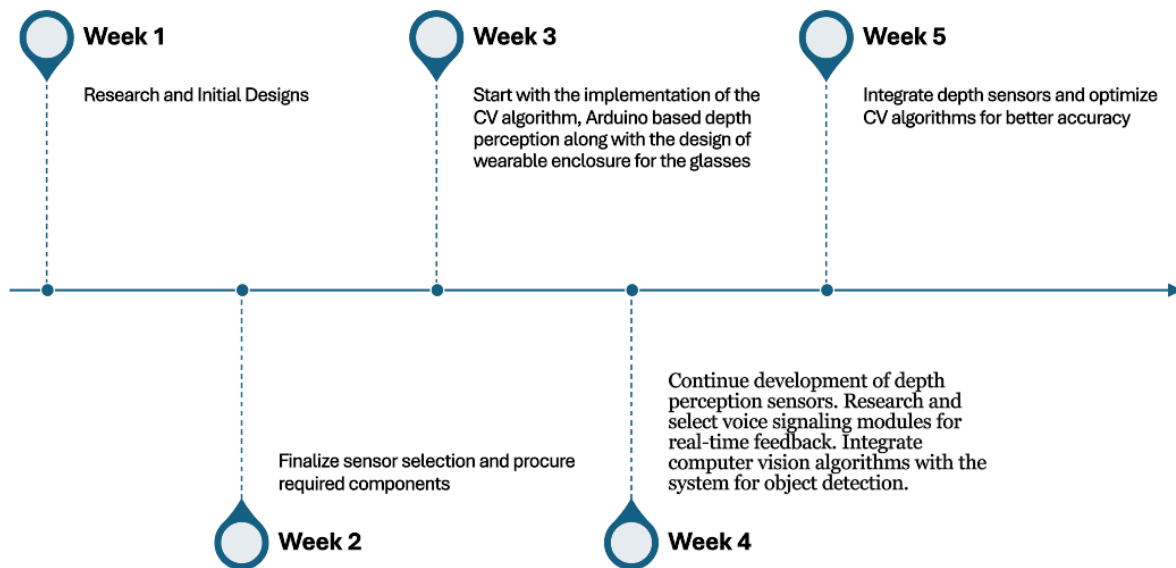
- Commence development of Arduino-based depth perception sensors (if applicable).
- Start designing the wearable enclosure for the glasses.
- Begin implementation of computer vision algorithms for object detection.

Week 4 :

- Continue development of depth perception sensors.
- Research and select voice signaling modules for real-time feedback.
- Integrate computer vision algorithms with the system for object detection.

Week 5 :

- Integrate depth perception sensors with Arduino microcontrollers.
- Conduct testing of basic functionalities (depth perception, object detection).
- Refine and optimize computer vision algorithms for improved accuracy.



Materials Needed:

Both options require

1. LiDAR or a camera module
2. Arduino microcontrollers
3. A wearable enclosure, voice signaling module
4. Power source, and wiring.

Option 1 emphasizes a camera module, while Option 2 focuses on Arduino-based depth perception sensors.

Methodology:

Two options are proposed for depth perception:

1. Camera-Based Depth Perception: This option replaces LiDAR sensors with a camera module, utilizing computer vision algorithms (stereo vision or structure from motion) for affordable depth estimation.
2. Arduino-Based Depth Perception Sensor: This option employs Arduino-based depth perception sensors like time-of-flight or ultrasonic sensors for distance measurement, providing an alternative to LiDAR.

Unique-Features:

The system offers flexibility with two depth perception options, enhancing safety by alerting users to nearby objects. Its affordability is emphasized through the use of off-the-shelf components and open-source software. The glasses are customizable, allowing personalized adjustments in voice signaling and object detection settings. They can also be used as a guide for the directions to the destination without anyone's help or memorizing the way. Basically, this

device will provide full independence to a impaired person in terms of mobility from source to destination.

Potential Applications:

The glasses find applications in enhanced navigation through crowded spaces, serving as an accessibility aid, and as an educational tool for spatial awareness training for individuals with visual impairments.

